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Virtualization:-

Virtualization is a technology that allow us to create multiple virtual computer on one physical computer. These virtual computers are called virtual machine. Each virtual machine can have its own operating system and applications. A hypervisor manage the resources and allow multiple VMs to share the same physical hardware.

Benefits of virtualization:-

1) Better use of hardware resources.
2) Reduced hardware and electricity cost.
3) Easy to create or remove virtual machine.
4) Supports cloud computing services.

Core architecture of virtualization:-

The architecture of virtualization has three main components:-

Physical Hardware:- It is the actual computer or server that provide CPU, RAM, storage and network resources.

Hypervisor:- It is software that creates and manage VM. It distributes hardware resources among VMs and maintain their separation.

Virtual machine:- These are the virtual computers that run their own operating systems and application.

Working of virtualization:-

Virtualization works through the following steps:-

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Physical server:- The computer provides CPU, RAM, storage and network resources.

Install hypervisor:- The hypervisor is installed directly on hardware or an an existing operating system.

Create VM:- The hypervisor creates multiple VMs.

Allocate resources:- CPU, RAM and storage are assigned to each VM.

Run application:- Each VM runs its own operation system and application.

A virtual machine behave like an independent computer, even though it share hardware with other VMs.

Types of hypervisors:- Hypervisor is software that creates and manages VMs. These are two main types:-

1) **Bare-metal hypervisor:-** Bare-metal hypervisor run directly on physical hardware with requiring a separate hosting operating system underneath it. **e.g:-** VMware ESXI, Microsoft hyper-v.

Features:-

1) Provide high performance.
2) Manages hardware resources directly.
3) Commonly used in cloud systems and data centers.

2) **Hosted hypervisor:-** Hosted hypervisor runs on an existing operating system. such as Windows or MacOS.

Features:-

1) Easy to install and use.
2) Suitable for learning and testing.

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3) May have lower performance than type 1.

Virtual Machine:- Virtual machine are often called cloud instances when they are provided as computing resources by a cloud platform.

i) VM behaves like an independent computer.

2) Each VM has its own operating system and application.

3) Physical computer are called host and VM are called guest.

Types of Virtualization:- The six major types of virtualization:

1) **Application Virtualization:-** Application virtualization allows user to access applications without installing them directly on their own devices. The application may run on a remotely remote server. **e.g:-** Citrix virtual apps allow user to access applications.

2) **Network Virtualization:-** Network virtualization creates and manages virtual networks through software instead of relying entirely on physical network arrangement. **e.g:-** AWS VPC allow user to create and manage virtual networks in cloud.

3) **Desktop Virtualization:-** Desktop virtualization allows a desktop operating system to run on central server. Users can access their desktops remotely through different devices. **e.g:-** Amazon workspace provide cloud-hosted desktops.

4) **Storage Virtualization:-** Storage virtualization combine multiple physical storage devices into one logical storage pool. This makes storage easier to manage. **e.g:-** Storage area network combines storage resources into centrally managed system.

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5) **Server Virtualization:-** Server virtualization divides one physical server into multiple virtual server. Each virtual server can run its own operating system and application. **e.g:-** Single physical server can run a linux web server and windows database server using separate VMs.

6) **Data Virtualization:-** Data virtualization provides a single view of data from different sources without requiring all the data to be moved to one location **e.g:-** Dashboard can display information from a database and a cloud data source together.

Virtualization vs. Cloud computing:-

Virtualization:- It is a technology that creates virtual versions of physical computing resources. It allow multiple VMs to share one physical server which improve hardware utilization and reduce cost.

Cloud Computing:- It provides computing services such as servers, storage and networking over a network usually internet.

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Kernel:- Kernel is the core part of an operating system It act as a bridge between computer hardware and software application. Kernel manage important computer resources such as CPU, memory and Input/output devices.

Types of kernel:-

1) **Monolithic kernel:-** Monolithic Kernel provides most OS services inside kernel space. This make it fast but a failure in one component may effect other parts of system.

2) **MicroKernel:-** Microkernel keeps only essential services inside the kernel while many other services run in user space. This improve reliability and isolation but communication between component can add overhead.

3) **Hybrid kernel:-** Hybrid kernel combines ideas from monolithic and microkernel designs. It keeps some services in the kernel for performance while using separation for other components.

4) **Nano kernel:-** Nanokernel is an extremely small kernel that provide basic hardware abstraction. Most other operating system services are handled outside it.

Operating system based virtualization:-

Operating system based virtualization is also called Containerization. It allow multiple isolated environment called container to run on the same operating system kernel.

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Traditional virtualization architecture:- In traditional virtualization, multiple virtual machines run on one physical computer. Each VM has its own operating system and application. Main components are:-

Virtual machine:- Each VM provide an isolated environment with its own operating system.

Virtual machine management layer:- Manage VMs, allocates resources and maintains isolation.

Host operating system:- Provide the environment in which the virtualization software runs.

Hypervisor:- Create and manages VMs and distributes physical resources among them.

Hardware:- The physical computer containing the CPU, memory and storage, input/output devices.

How OS-based virtualization works:- OS-based virtualization work by allowing multiple containers to share the same OS kernel.

1) **Shared Kernel:-** All containers use the host OS kernel instead of having separate kernel.

2) **Isolation:-** The kernel keeps containers separate using feature called namespace. These help isolate processes, network and file system.

3) **Resources management:-** Control groups manage and limit the CPU, memory, disk input/output and network resources used by each container.

4) **Resources limit:-** The system controls the resources available to each container according to its limit.

5) **Access control:-** Container cannot access resources outside its

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environment unless permission is given.

6) **Low overhead:-** Containers need fewer resources than traditional VMs because they do not require a complete separate operating system.

Operating system based services:- Main services are:-

1) **Backup and recovery:-** The host operating system and backup tools can help create backups and restore system or virtual machine when needed.

2) **Security management:-** Security can be managed by configuring firewalls, installing antivirus software, and applying other security setting.

3) **Integration with directory services:-** The host OS can connect with directory services such as active directory to manage user and groups centrally.

Operating System based operations:- OS support several operations

1) **Using hardware capabilities:-** Containers can use available resources such as the CPU and network connection.

2) **Using Connected devices:-** The host OS can interact with devices such as webcams, printers, keyboards and scanner.

3) **Managing files and data:-** The host OS can read and write data in files, folders and network shares.

Feature of OS-based virtualization:-

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1) **Resource isolation:-** Each container receives allocated resources such as CPU time, memory and Input/output capacity

2) **Portability:-** Containers can be moved between compatible environment with fewer changes to the application.

3) **Scalability:-** Containers can be added or removed according to application requirements and workload.

4) **Security:-** Containers isolate application from one another, helping limit unwanted access. Because they share same kernel they do not provide complete security isolation.

5) **Easy management:-** Containers can be started stopped and monitored using commands and container management tools

Pros of OS-based virtualization:-

1) **Resource Efficiency:-** Container use fewer resources because they share the host OS kernel.

2) **High Scalability:-** Containers can be quickly increased or decreased according to demand.

3) **Easy Management:-** Containers can be managed using commands and tool making deployement and maintenance easier.

4) **Reduced cost:-** Fewer resources and less infrastructure may be required compared with running many full virtual machines

5) **Faster deployement:-** Applications can be started and update quickly using containers.

Cons of OS-based virtualization:-

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1) **Security**:- Since containers share the host OS kernel, a kernel vulnerability or security based may affect multiple containers.

2) **Limited isolation**:- Containers do not provide fully isolation as fully separate VM in every situation. They may also compete for resources.

3) **Complexity**:- Setting up and managing containers securely can require specialized knowledge.

4) **Limited hardware access**:- Some application may not able to access hardware directly or may require special configuration to use certain devices

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System models:- There are two types of system model:- a non-virtual model and virtual model In non-virtual model the operating system work directly with physical hardware. In virtual machine system a VMM creates multiple virtual environment, allowing several operating system to run on same physical machine

Implementation of VMMs:- VMMs can be implemented in different ways:-

Type 0 hypervisor:- Type 0 hypervisor use hardware or firmware features to create and manage VMs **eg:-** including Oracle LDOMs.

Type 1 hypervisor:- Type 1 hypervisor run directly on hardware or provide operating system like virtualization **eg:-** Xen server, Hyper-V.

Type 2 hypervisor:- Type 2 hypervisor run as applications on standard operating system such as VMware workstation, Parallel desktop

Implementation of VMs (cont.):- There are also other virtualization approaches **Paravirtualization** modifies the guest operating system so it can cooperate with VMM for better performance.

Programming-environment virtualization:- used by Java and Microsoft .NET, provides a special environment for programs.

Emulation:- Emulation allow software designed for one hardware architecture to run on another.

Application containment:- Application containment such as Solaris Zones, BSD Jails separate application to improve security and management without providing full hardware virtualization.

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History:-

Virtual machine first appeared in IBM mainframes system in 1972 allowing multiple user to share a batch-oriented system. Virtualization helped the technology spread beyond IBM. In late 1990s faster intel processors encouraged researches to explore virtualization on general purpose PCs and technologies such as Xen and VMware help expand it use across different operating system and CPU.

Benefits and features:- Virtual machines are isolated from host and each other, reducing the spread of viruses. They can be suspended, moved, copied and resumed. Snapshots restore on earlier state, while cloning create a copy of VM

Benefits and features (cont.):- Templating creates multiple VMs from one template. Live migration moves a running VM to another host without interrupting user's. These feature help support cloud computing.

Building blocks:- Creating an exact copy of hardware is difficult but modern CPU support make virtualization easier A VCPU store each guest CPU state which the VMM save and loads when needed.

Building blocks — Trap and emulate:- Guest code runs in user mode while the guest OS expects Kernel mode. The VMM uses virtual user and Kernel modes both running in real user mode. When a guest attempts a privileged operation the VMM handles it safely.

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Trap and emulate:-

Trap and emulate is a technique in which the guest OS runs in virtual user mode and virtual kernel mode. but both operate in real user mode. When the guest attempts a privileged instruction the CPU generates a trap. and the VMM take control to perform operation. After completing the task the VMM returns control to the guest OS. This method is safe but frequent traps can reduce performance So, modern CPU provide hardware support to improve virtualization.

Trap and emulate implementation:-

In trap and emulate the guest OS attempts to execute a privileged instruction which generates a trap because the guest cannot execute it directly. The VMM receives the trap examines the instruction and perform the required operation on behalf of guest. After completing the operation the VMM returns control to guest OS. This process allows the guest to perform privileged task safely.

Binary translation:-

Binary translation is a technique used when some CPU instructions cannot be handled correctly by Trap and emulate. The VMM check the instruction and changes special or problematic instructions into safe

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equivalent instruction. Normal instructions run directly while translated instructions are saved and reused to improve speed. According this Starting Windows XP as a guest needed about 950,000 translations took around 3 seconds and caused a 5% slowdown compared with running it directly on computer.

Binary translation implementation:-

The VMM checks the instructions that the guest OS wants to run. Normal instruction run directly on CPU, while special or problematic instructions are changed into equivalent instructions. These new instructions perform the required the task safely. This method helps run older CPUs that do not properly support the trap and emulate method.

Type 0 hypervisor:-

Type 0 hypervisor uses firmware to manage virtual machines directly on hardware. It is also called partitions or domains, depending on the hardware manufacter. Each guest usually gets its own hardware resources and its OS does not need to be modified. If there are not enough input/output devices for every guest a control partition manages shared devices. Type 0 systems also allow a guest to run its own hypervisor and create more virtual machine.

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Type 1 hypervisor:-

Type 1 hypervisor runs directly on physical hardware and is commonly used in company data centers. It creates and manages multiple guest OS allow organizations to run more applications on fewer servers. It manages CPU memory and hardware resources. It also allows users to take snapshots, clone virtual machine and move them between server. Some type 1 hypervisor run also on type 0 hypervisor.

Type 1 hypervisor is a general purpose OS that also provide virtualization. E.g:- Red-hat enterprises linux with KVM, Windows with hyper-V. These system perform normal OS task as well as hypervisor task. They usually have fewer virtualization features than dedicated type 1 hypervisor. They treat guest OS like normal processes but handle special instruction differently.

Type-2 hypervisor:-

Type-2 hypervisor run as a normal application on host OS. which manages it like any other process. The host OS may not even realize that the application is running VM. Type 2 hypervisor usually have lower performance because they cannot use some hardware directly. However they do not require changes to host OS.

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Paravirtualization :-

Paravirtualization is different from normal virtualization because the VMM does not provide an exact copy of real hardware. Instead the VMM provides special services that the guest OS must be modified to use. This can improve performance although it is becoming less necessary as hardware support for virtualization improves. Xen is an example of Para virtualization. It provides simple device interfaces, efficient I/O and better communication between the guest and VMM. For device I/O, the guest and VMM share a circular buffer through shared memory.

Nested page tables :-

Nested page tables help the VMM manage memory for VM. Each guest OS maintains its own page table to translate virtual address into physical address. While the VMM maintain separate nested page tables for each guest. When a guest runs on CPU, the VMM makes that guests nested page table active. If the guest changes its page tables the VMM makes the same changes in the nested page tables and its own page table. However this method can cause more TLB misses which can slow down performance.

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Paravirtualization:- Paravirtualization modifies the guest OS so it can cooperate with hypervisor. This cooperation can improve performance especially for input/output and memory management. Guests can use a hypercall to request services directly from hypervisor. **e.g:-** Xen

Programming Environment Virtualization:- Programming environment virtualization provides a special environment for running programs rather than copying physical hardware. The Java virtual machine is an example that allow Java programs to run on different system that support it. This makes program more portable across OS and hardware architecture

Emulation:- Emulation allows software designed for one CPU architecture to run on different architecture. It translates the guest instructions into instruction that the host computer can execute. It is useful for running old applications and games on newer system but it can be slower because instruction must be translated.

Application Containment:- Application containment separate application into isolated environment while sharing the same OS kernel. Each environment can have its own application, network setting, user account, and allocated resources. This improve security and management while using fewer resources than full virtual machine. **E.g:-** Oracle Solaris Zone.

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Virtualization and OS components:- Virtualization affects CPU Scheduling, memory management, Input/output, storage and migration. The hypervisor must manage physical resources even when each guest behaves as if it has own hardware. Efficient resource management help maintain performance while multiple VMs run on one computer.

CPU scheduling:- The hypervisor decides how physical CPU time is shared among virtual CPUs belonging to different guests. If there are more virtual CPUs than physical CPUs, CPUs overcommitment occurs and VMs must share processing time. This can reduce performance and delay application resources. It may also affect time keeping inside guest OS.

Memory management:- The hypervisor manage memory when multiple VMs compete for limited physical RAM. Double paging moves page to storage even when a guest believes they are in physical memory. A balloon memory manager ask the guest to release memory, while deduplication allows identical memory pages to be shared between VMs reducing memory usage.

Input/output :- The hypervisor provide virtual devices and drives so guest can access disks, networks and other devices. I/O management is complex so hardware support can help provide direct device access. DMA pass-through and direct interrupt delivery. For networking the hypervisor can connect guests directly to

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a network or use Network address translation (NAT) to translate their network addresses.

Storage Management:- Virtual machines store OS, applications and data in virtual disk files or other storage devices. Type 1 hypervisors may store virtual disks in their own file system. while Type 2 hypervisor usually store them as file managed by host OS. Copying or moving these files can help create or move VMs. Physical-to-virtual converts physical disks to virtual format while virtual-to-physical convert virtual disk to physical format.

Live migration:- Live migration moves a running VM from one physical server to another without interrupting user access. The source hypervisor copies the guest memory to the target and repeats the process for memory pages that change. It then briefly pauses the guest to transfer the final CPU state and remaining changes. The target starts the VM and the source removes its copy after confirmation.

Example - VMware:- VMware workstation is a type 2 hypervisor that runs on an existing OS. It allows multiple guest OS such as Windows and Linux to run at same time. Each guest receive virtual hardware, including CPUs, memory, disk and network interface. Virtual disk can be stored as files on host computer.